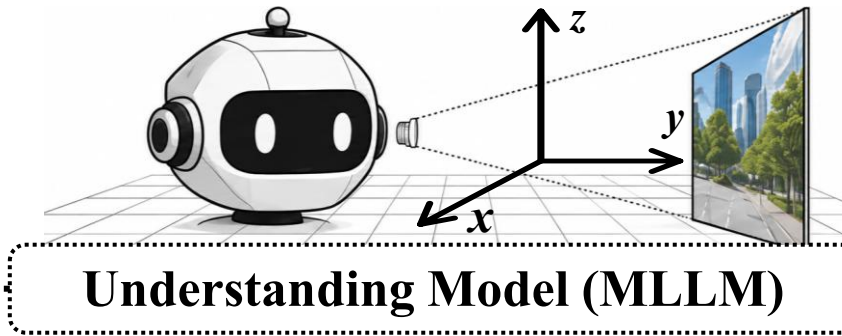


# Spatial Memory Intelligence



## Memory Update

## Memory Retrieve

### 1. Spatial Clustering

Newly Generated Chunk

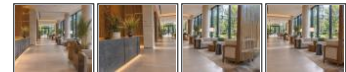


Similar Cluster?

Yes

Assign to Matched Cluster

Cluster 1



Cluster 2



Cluster 3



Create New Cluster

New Cluster

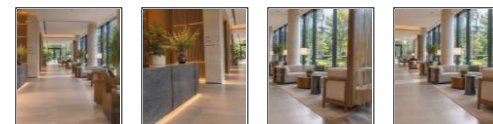


Read

Write

### 2. Within-Cluster Sparsification

Cluster



If Cluster Budget Is Exceeded

Sparsify (Remove Redundancy)



Retained Compact Memory

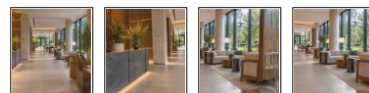


Read

Write

Persistent Spatial Memory ( $\mathcal{M}_k$ )

Cluster 1



Cluster 2



Cluster 3



### 3. Action-Aware Retrieval

Recent Context



+ Current Action

Move Forward

Turn Left

Selected Memory



Forward



Left

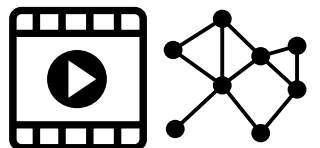
Read

Selected Memory

Recent Context

Current Action

Video World Model



Generated Chunk

### 4. Reliability-Aware Filtering



Drift



Discard

Reliable

